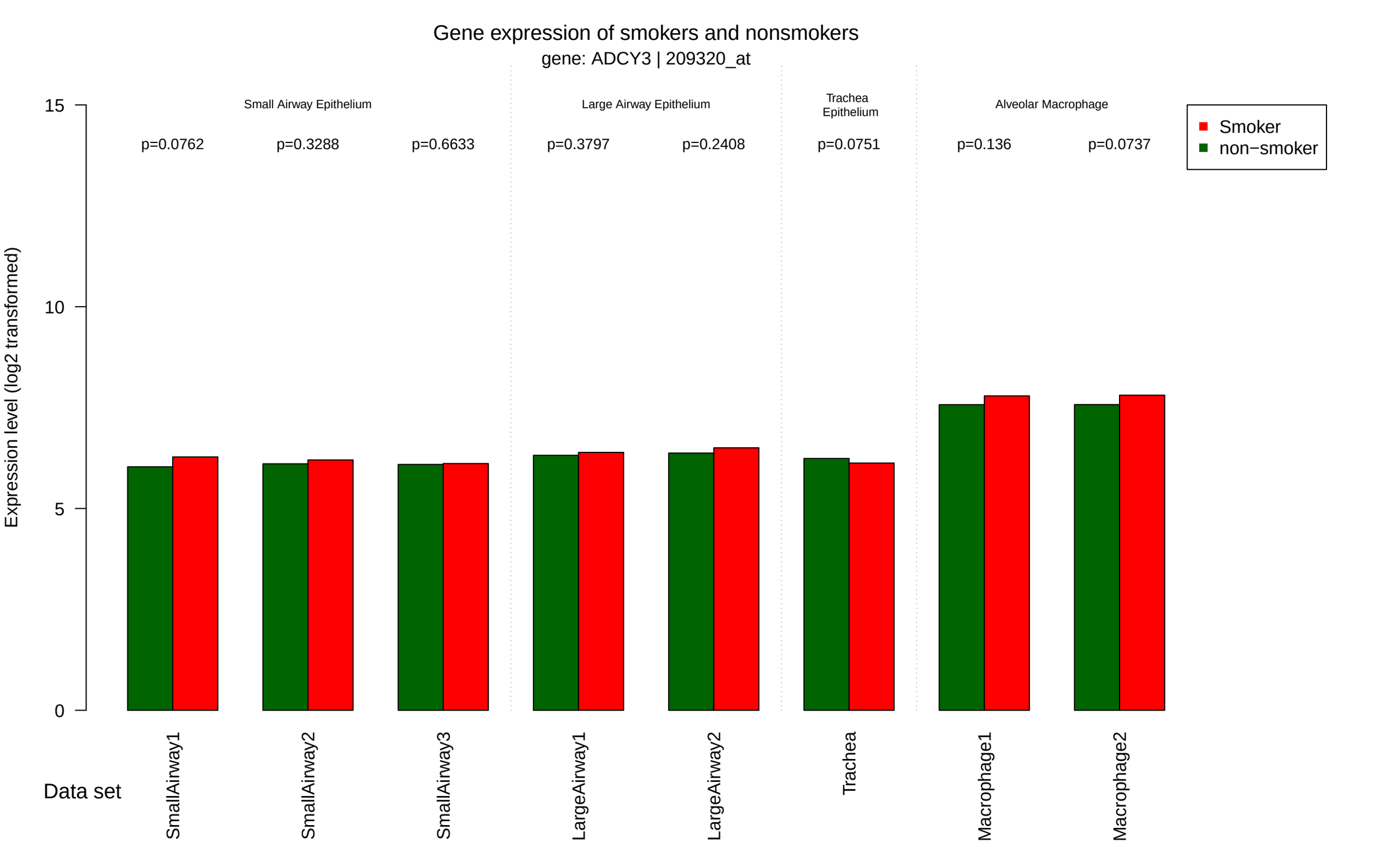
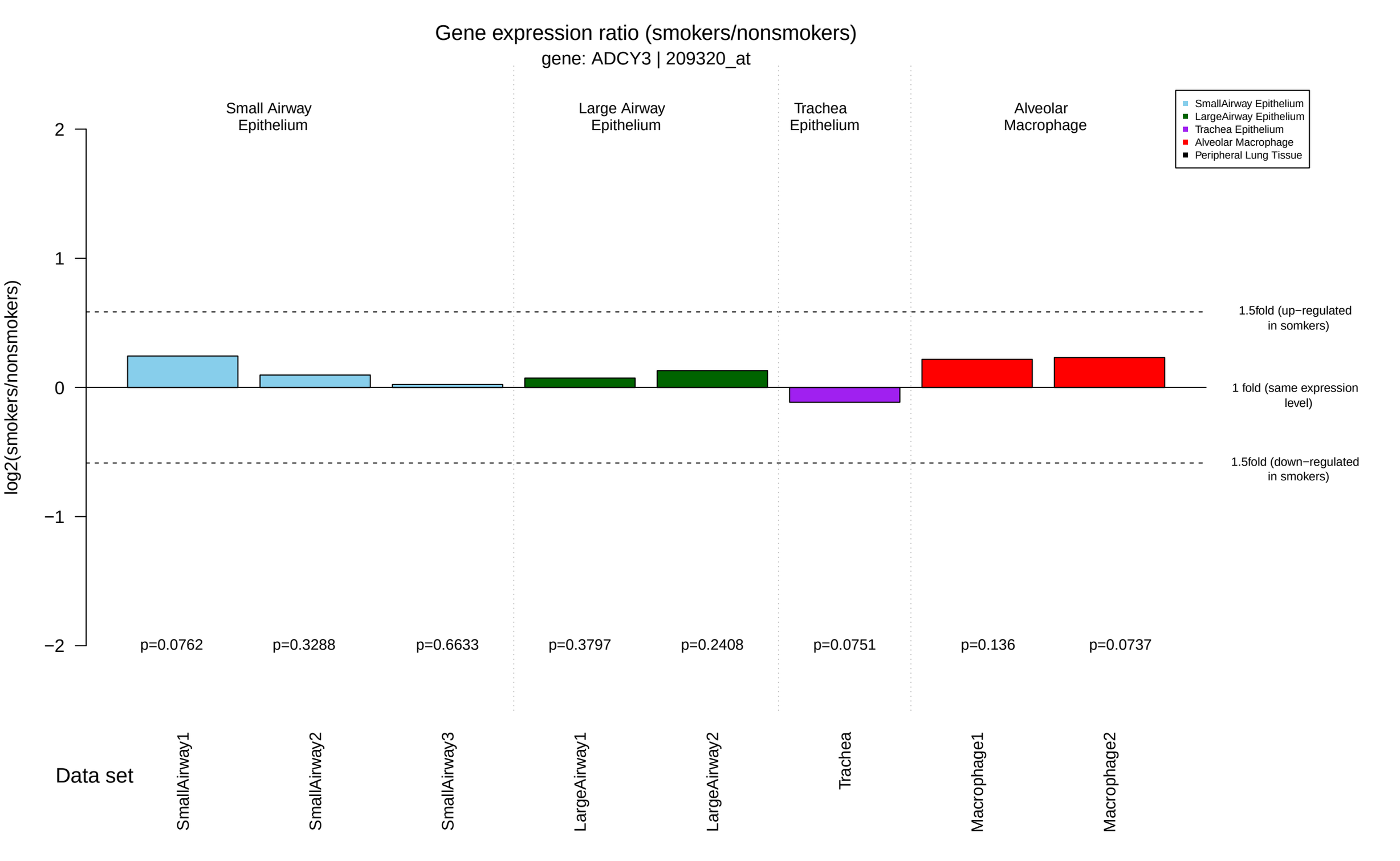
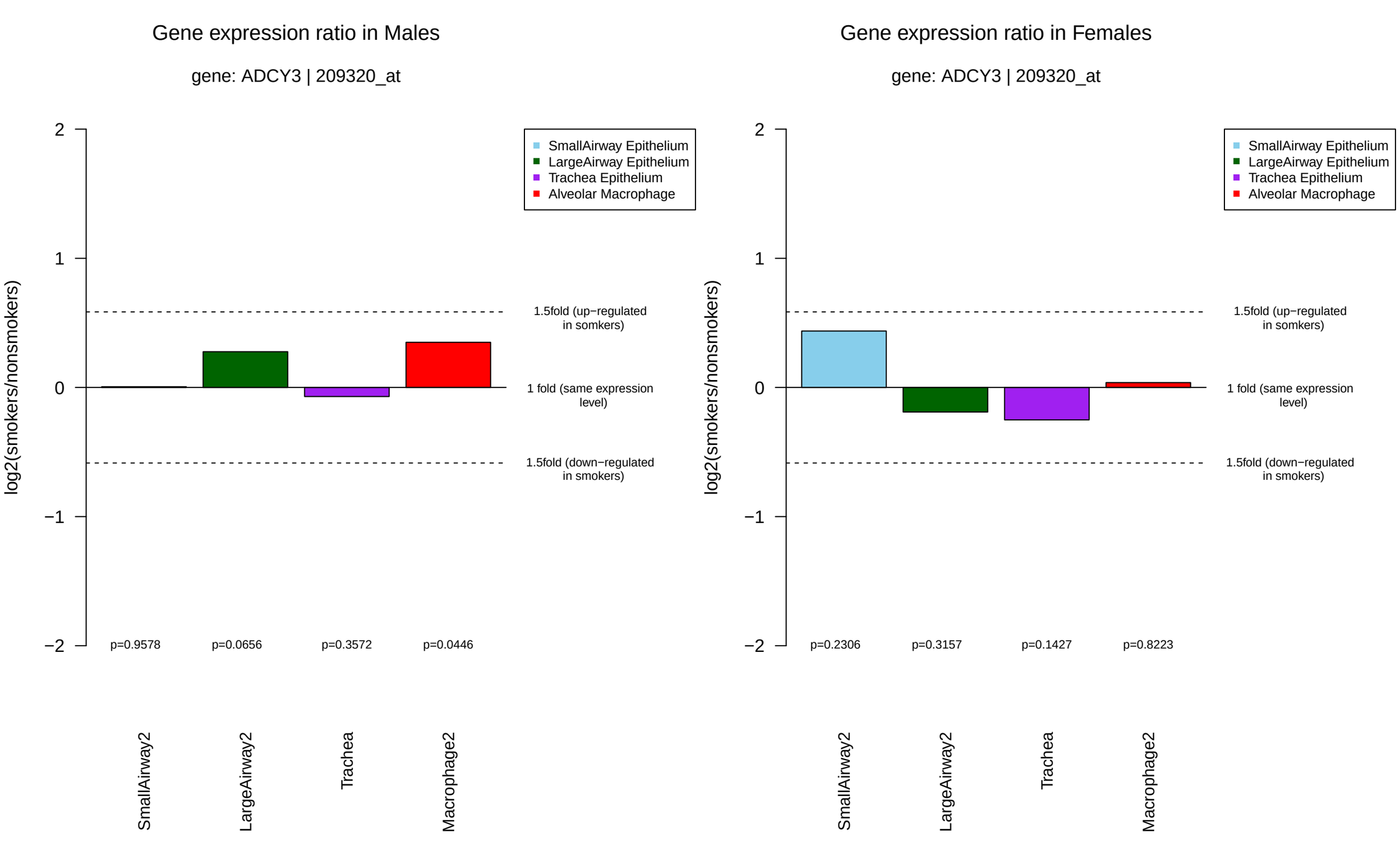
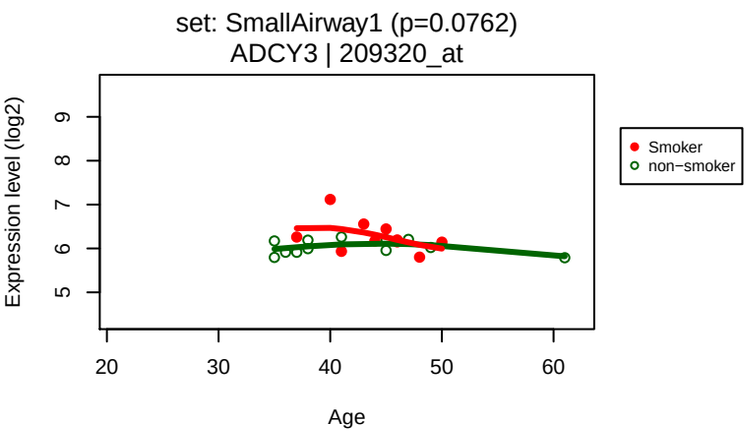
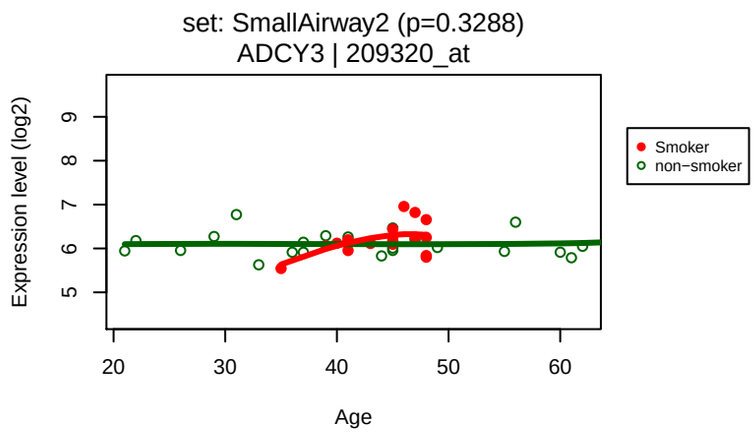
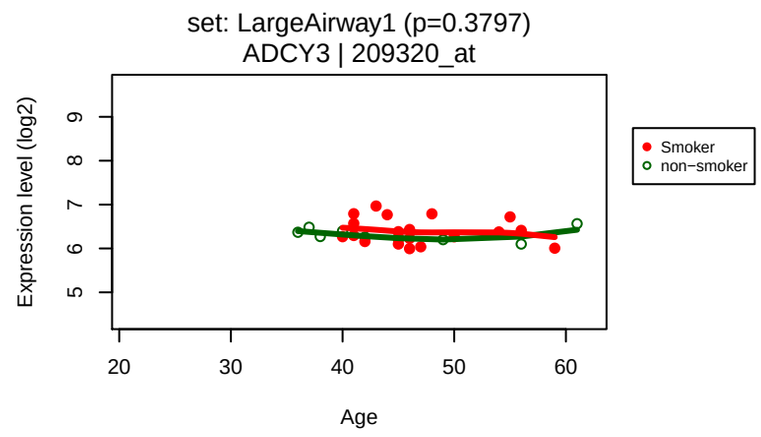

Gene expression of smokers and nonsmokers
gene: ADCY3 | 209320_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0762
p=0.3288
p=0.6633
p=0.3797
p=0.2408
p=0.0751
p=0.136
p=0.0737
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

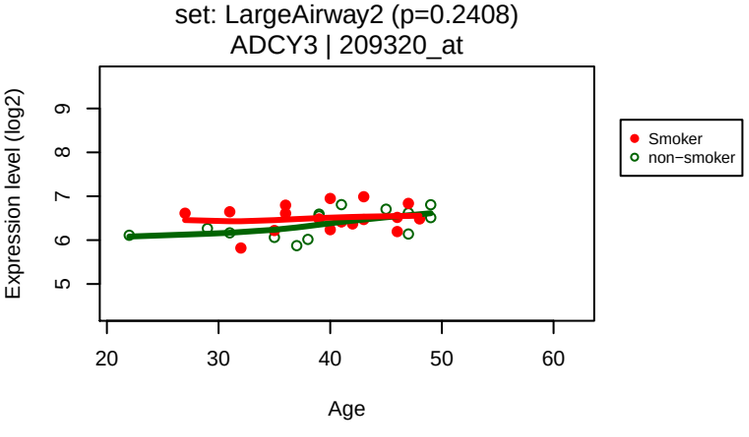

Gene expression ratio (smokers/nonsmokers)
gene: ADCY3 | 209320_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0762
p=0.3288
p=0.6633
p=0.3797
p=0.2408
p=0.0751
p=0.136
p=0.0737
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

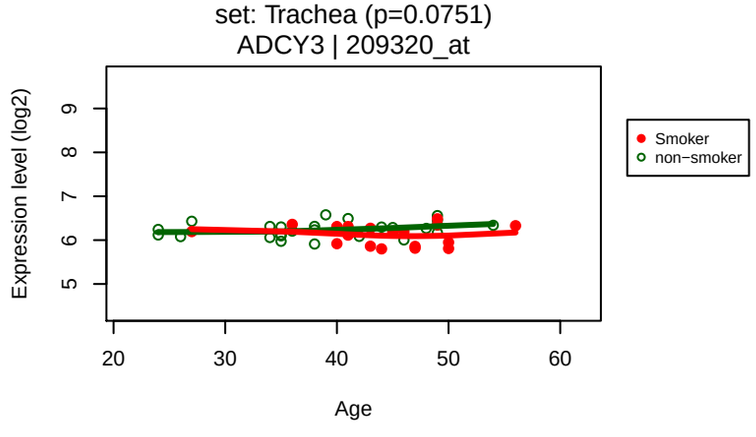

Gene expression ratio in Males
gene: ADCY3 | 209320_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9578
p=0.0656
p=0.3572
p=0.0446
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY3 | 209320_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2306
p=0.3157
p=0.1427
p=0.8223
SmallAirway2
LargeAirway2
Trachea
Macrophage2

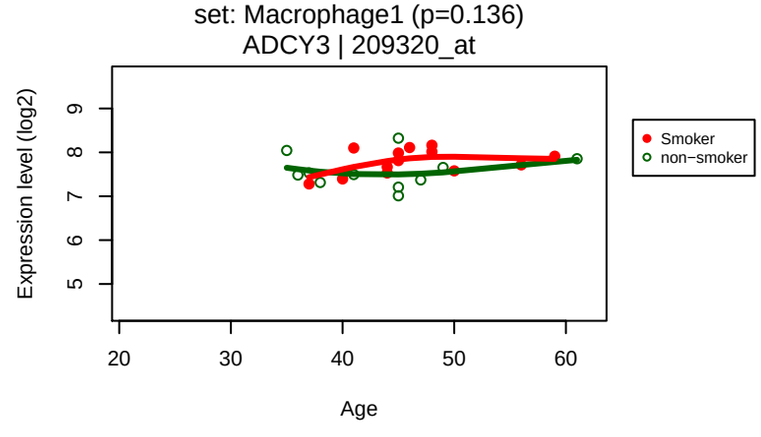

set: SmallAirway1 (p=0.0762)
ADCY3 | 209320_at
Expression level (log2)
Age
Smoker
non−smoker

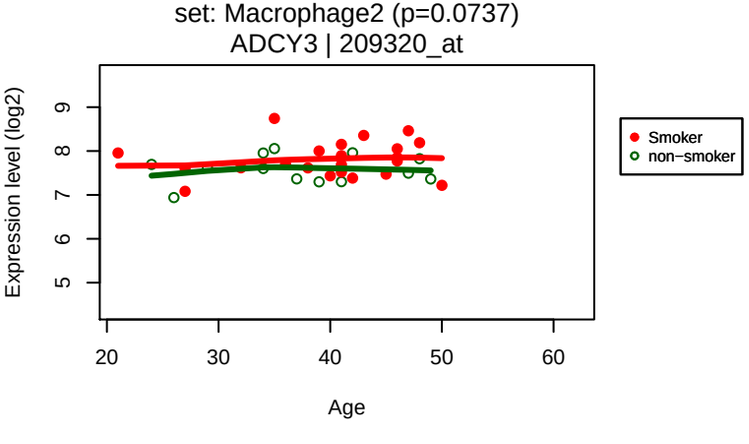

set: SmallAirway2 (p=0.3288)
ADCY3 | 209320_at
Expression level (log2)
Age
Smoker
non−smoker

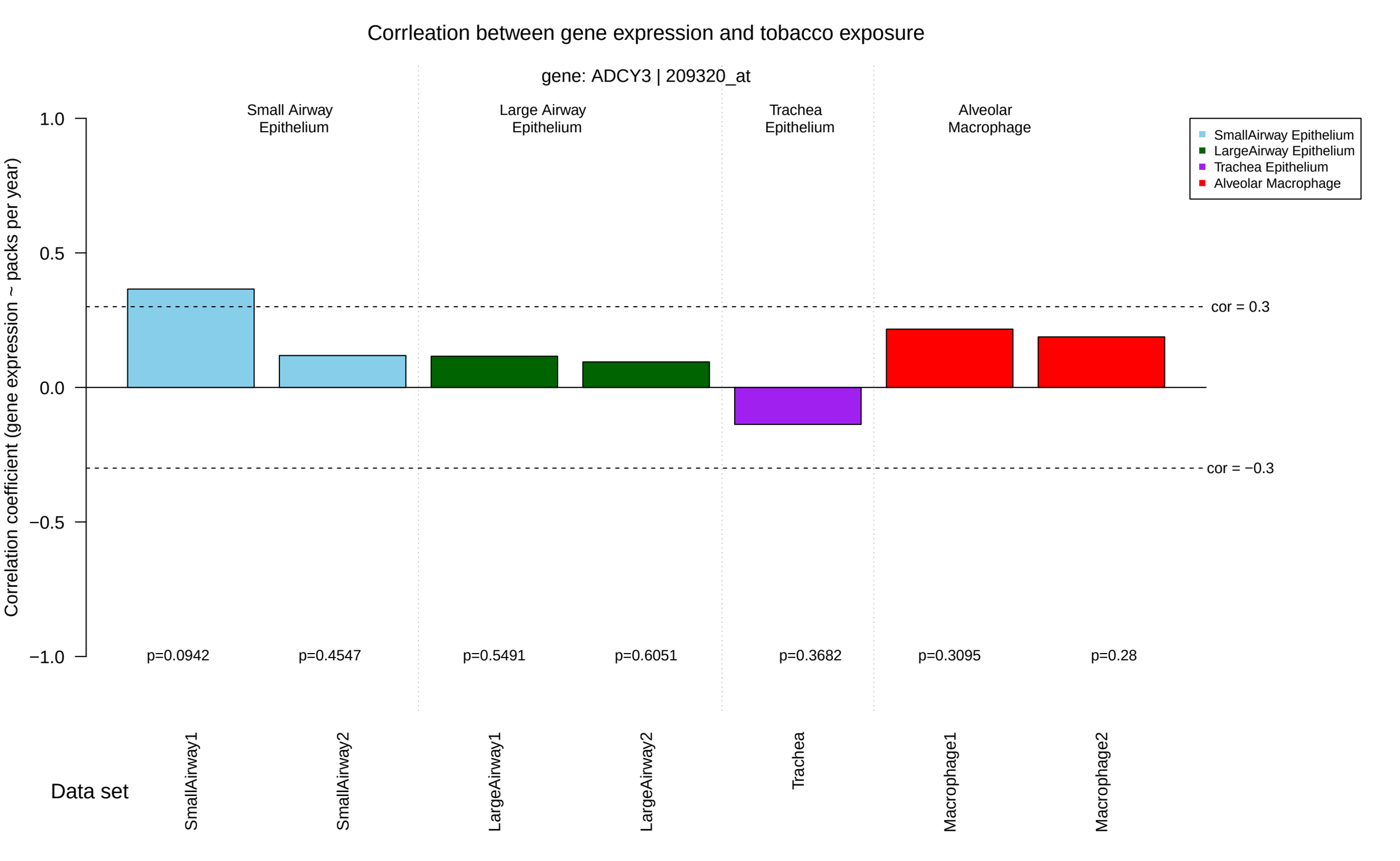

set: LargeAirway1 (p=0.3797)
ADCY3 | 209320_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2408)
ADCY3 | 209320_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0751)
ADCY3 | 209320_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.136)
ADCY3 | 209320_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0737)
ADCY3 | 209320_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY3 | 209320_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0942
p=0.4547
p=0.5491
p=0.6051
p=0.3682
p=0.3095
p=0.28
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2